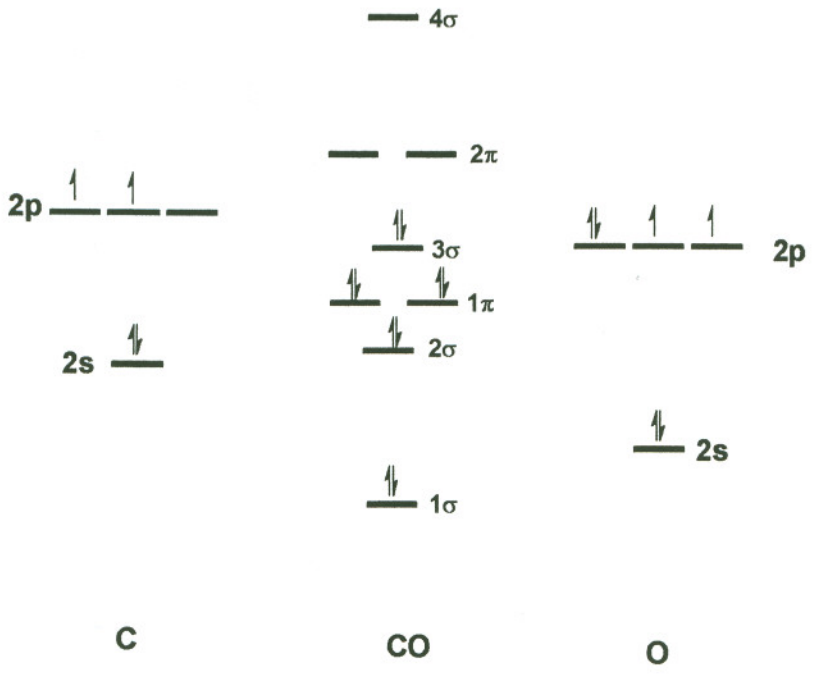
Inorganic Cume October 19,2001

1) Give a simple yet convincing argument to justify the linearity of NCS^{1-} and the bent geometry for the NO_2^{1-} ions.

2) Thiocyanate acts as an ambidentate ligand, i.e. it can bond to a Lewis acid using either of its external atoms. Give <u>two</u> different bonding arguments you would use to predict the preferred mode of bonding to H^{1+}.

3) Use the MO correlation diagram below to justify the characteristics observed for metal – CO bonding.

   a) Why is M-C bonding preferred over M-O attachement for CO ligands?
   b) Why is CO a good ligand to stabilize low oxidation states?
   c) Why is ν(CO) shifted to lower frequency in carbonyl complexes compared to free CO?



4) Use a simplified MO correlation diagram (one orbital/atom) to compare the O....H - O hydrogen bonding interaction to the B – H – B bonding in boranes.